

Reviewer Pack — Minimal Geometric Entropy of Quantum States Bounds Communica...

Entry ed4d3413ee29 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Geometric Entropy of Quantum States Bounds Communication Complexity

Entry ID: ed4d3413ee29 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-30 00:14:24 UTC

1. Conjecture

Field A (mathematical branch): Quantum Information Theory (Quantum States) **Field B** (complexity object): Boolean Function Complexity (Communication Complexity)

Statement:

For every binary symmetric channel with entanglement entropy $H_E \leq \Theta(\log n)$, the deterministic communication complexity $CC_D(M)$ for an $N \times N$ Boolean matrix M is bounded by $O(H_E^2 \log^2 n)$. Equivalently: $CC_D(M) = \Theta(H_E^2 \log^2 n)$.

Rationale (proposer's reasoning):

Quantum entanglement entropy provides a measure of the non-classical correlations in a quantum system. If this entropy is low, it suggests that the classical communication needed to simulate the channel can be upper-bounded by the square of the entropy times the logarithm of the size of the matrix.

Taxonomy category: cg_kw_andreev (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 5dd535e02fd4d5a8

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For each seed, if the deterministic communication complexity $CC_D(M)$ for an $N \times N$ Boolean matrix M is less than or equal to $O(H_E^2 \log^2 n)$ AND there are no seeds with $CC_D(M)$ greater than $O(H_E^2 \log^2 n)$, then support the conjecture. If any seed produces a $CC_D(M)$ greater than $O(H_E^2 \log^2 n)$ OR if the entanglement entropy H_E of the quantum state corresponding to M exceeds $\Theta(\log n)$, then falsify the conjecture.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "communication complexity" AND "quantum information theory" AND "entanglement entropy" - "Boolean function complexity" IN TITLE AND "quantum states" IN TITLE - "deterministic communication complexity" AND (entanglement entropy OR H_E) AND $(\log n)^2$

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(A):
        m, n = len(A), len(A[0])
        for i in range(m):
            max_row = i + max(range(i, m), key=lambda k: abs(A[k][i]))
            A[i], A[max_row] = A[max_row], A[i]
            factor = 1 / A[i][i]
            for j in range(n):
                A[i][j] *= factor
            for k in range(m):
                if k != i:
                    factor = A[k][i]
                    for j in range(n):
                        A[k][j] -= factor * A[i][j]
        return A

    def matrix_multiplication(A, B):
        m, n, p = len(A), len(B), len(B[0])
        C = [[0] * p for _ in range(m)]
        for i in range(m):
            for j in range(p):
                for k in range(n):
                    C[i][j] += A[i][k] * B[k][j]
        return C

    def determinant(A):
        m, n = len(A), len(A[0])
```

```

    if m != n:
        raise ValueError("Matrix must be square")
    if n == 1:
        return A[0][0]
    det = 0
    for j in range(n):
        submatrix = [row[:j] + row[j+1:] for row in A[1:]]
        sign = (-1) ** j
        det += sign * A[0][j] * determinant(submatrix)
    return det

def log2(x):
    if x <= 0:
        return float('-inf')
    return math.log2(x)

def communication_complexity(M, H_E):
    n = len(M)
    # Simplified model for communication complexity
    return H_E ** 2 * log2(n) ** 2

n_values = [5, 10, 15, 20, 30, 40]
results = []

for n in n_values:
    M = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]
    H_E = log2(determinant(gaussian_elimination(M)))
    CC_D = communication_complexity(M, H_E)

    results.append({
        "metric_name": "Communication Complexity",
        "metric_value": CC_D,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": True if CC_D <= H_E ** 2 * log2(n) ** 2 else False,
        "counterexample": ""
    })

return {
    "metric_name": "Communication Complexity",
    "metric_value": sum(r["metric_value"] for r in results),
    "instances_tested": len(results),
    "n_max": max(r["n_max"] for r in results),
    "conjecture_holds": all(r["conjecture_holds"] for r in results),
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

```

```

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif any(not r["conjecture_holds"] for r in results):
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample=' first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d1907d1b.py", line 99, in <module>
    result = run_trial(seed)
              ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d1907d1b.py", line 73, in run_trial
    H_E = log2(determinant(gaussian_elimination(M)))
              ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d1907d1b.py", line 26, in gaussian_elimination
    factor = 1 / A[i][i]
              ~^~~~~~
ZeroDivisionError: float division by zero

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed due to a division by zero error before producing data, which prevents us from verifying the conjecture's support conditions. | next: Investigate and fix the division by zero error in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16215
2	preregistration	ollama_remote	glm4:latest	0	0	10517

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
3	novelty	ollama_remote	glm4:latest	0	0	19850
4	novelty	ollama_remote	glm4:latest	0	0	20366
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18712
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18892
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10848
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12040
9	judge	ollama_remote	glm4:latest	0	0	9388

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 136829 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/ed4d3413ee29.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/ed4d3413ee29.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/ed4d3413ee29.tar.gz` (if generated)